
机器人学

第五章：机械臂运动学(下)

厦门大学信息学院

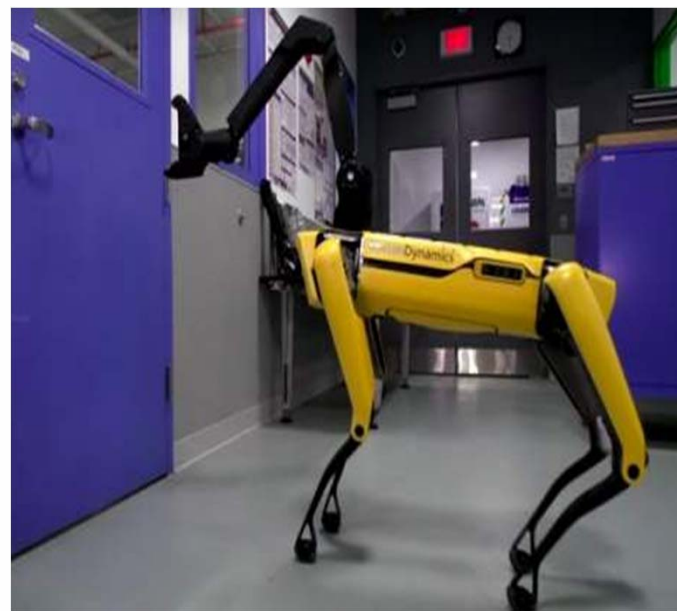
人工智能系

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**工业机器人抓紧产
线上的零部件**



**波士顿动力公司
SpotMini机器人**



协作机器人焊接工艺解决方案

00:00 / 01:36

倍速



引言

□ 手臂正运动学 Forward kinematics (FK)

给予 θ_i (可计算出 ${}^{i-1}T$) , 求得 $\{H\}$ 或 wP

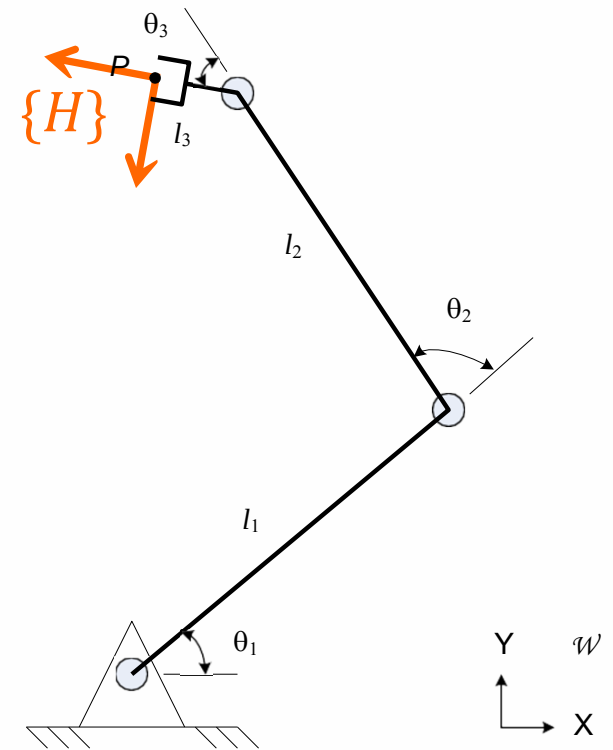
$${}^w_H T = f(\theta_1, \dots, \theta_i, \dots, \theta_n)$$

$${}^wP = {}^w_H T {}^H P$$

□ 手臂逆运动学 Inverse kinematics (IK)

给予 $\{H\}$ 或 wP , 求得 θ

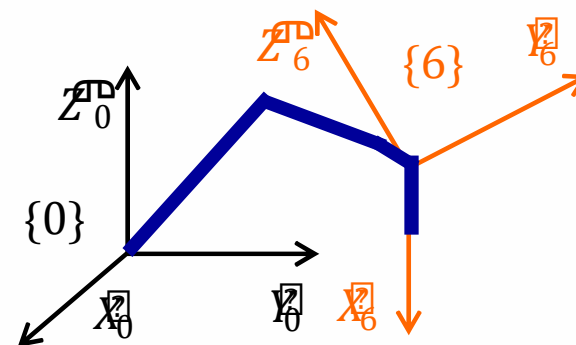
$$[\theta_1, \dots, \theta_i, \dots, \theta_n] = f^{-1}({}^w_H T)$$



求解概念 -1

□ 假设手臂有6 DOFs

- ◆ 6 个未知的joint angles (θ_i 或 d_i , $i = 1, \dots, 6$)



□ 在 wT_H 中取出含未知数的 0T_6 , 16个数字

$${}^0_6T = \begin{bmatrix} {}^0_6R_{3 \times 3} & {}^0P_{6\text{org}}_{3 \times 1} \\ 0 & 0 & 0 & 1 \end{bmatrix}_{4 \times 4} = \begin{bmatrix} | & | & | & | \\ {}^0\hat{X}_6 & {}^0\hat{Y}_6 & {}^0\hat{Z}_6 & {}^0P_{6\text{org}} \\ | & | & | & | \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

□ 求解

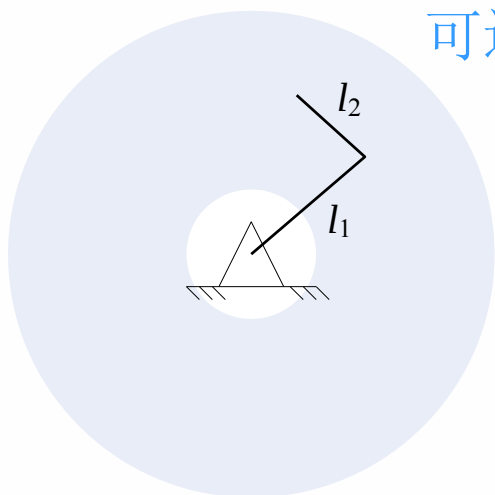
- ◆ 12个非线性超越方程式
- ◆ 6个未知数, 6个限制条件

求解概念 -2

- 可达工作空间
 - 手臂可以用一种以上的姿态到达的位置
- 灵活工作空间
 - 手臂可以用任何的姿态到达的位置

□ Ex: A RR manipulator

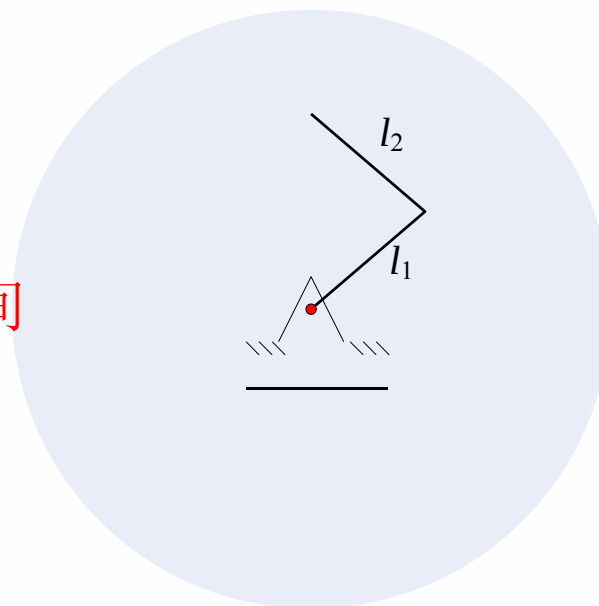
If $l_1 > l_2$



可达工作空间

灵活工作空间

If $l_1 = l_2$



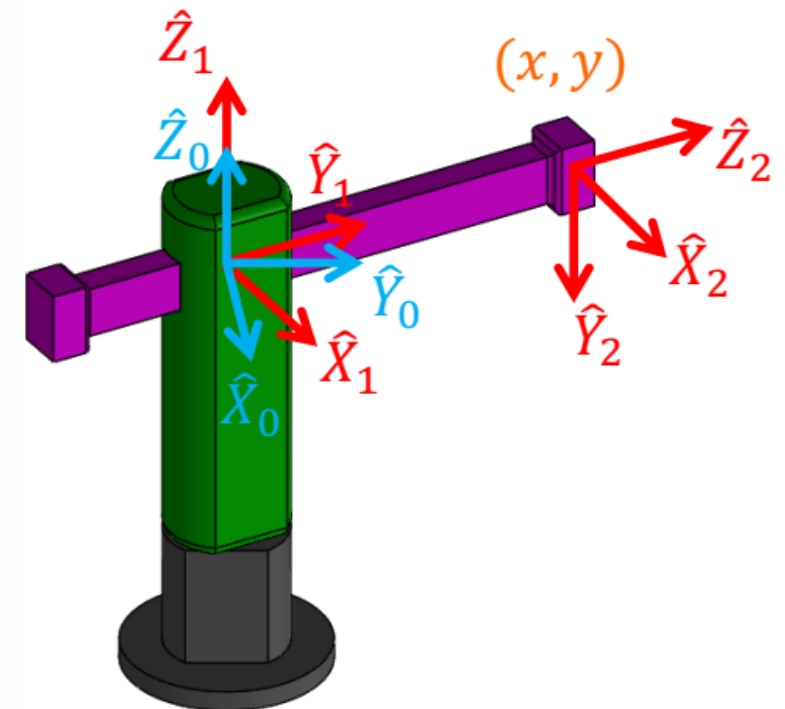
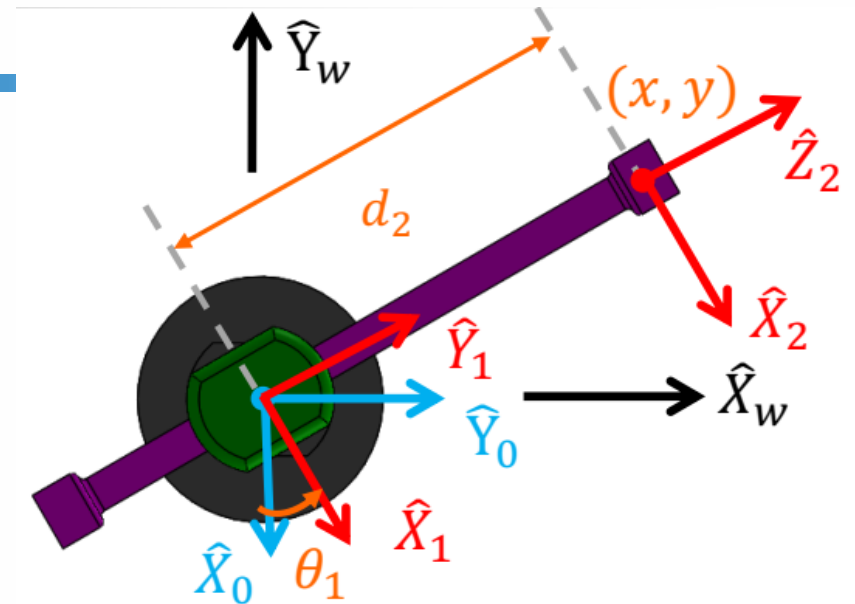
求解概念 -3

□ 子空间

- ◆ 手臂在定义头尾的 T 所能达到的变动范围

□ Ex: A RP manipulator

- ◆ 2- DOF, Variables: (x, y)



$${}^w_2T = \begin{bmatrix} \frac{y}{\sqrt{x^2 + y^2}} & 0 & \frac{x}{\sqrt{x^2 + y^2}} & x \\ \frac{-x}{\sqrt{x^2 + y^2}} & 0 & \frac{y}{\sqrt{x^2 + y^2}} & y \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

${}^0\hat{Z}_2$ ${}^0P_{2 ORG}$

多重解 -1

□ 解的数目

- ◆ 由于是非线性超越方程，6未知数6方程式不代表具有唯一解
- ◆ 是由joint数目和link参数所决定

Ex: A RRRRRR manipulator

a_i	解的数目
$a_1 = a_3 = a_5 = 0$	≤ 4
$a_3 = a_5 = 0$	≤ 8
$a_3 = 0$	≤ 16
$All\ a_i \neq 0$	≤ 16

多重解 -2

□ Ex: PUMA(6 rotational joints)

- ◆ 针对特定工作点，8组解

- ◆ 前3轴具有4种姿态

如右图所示

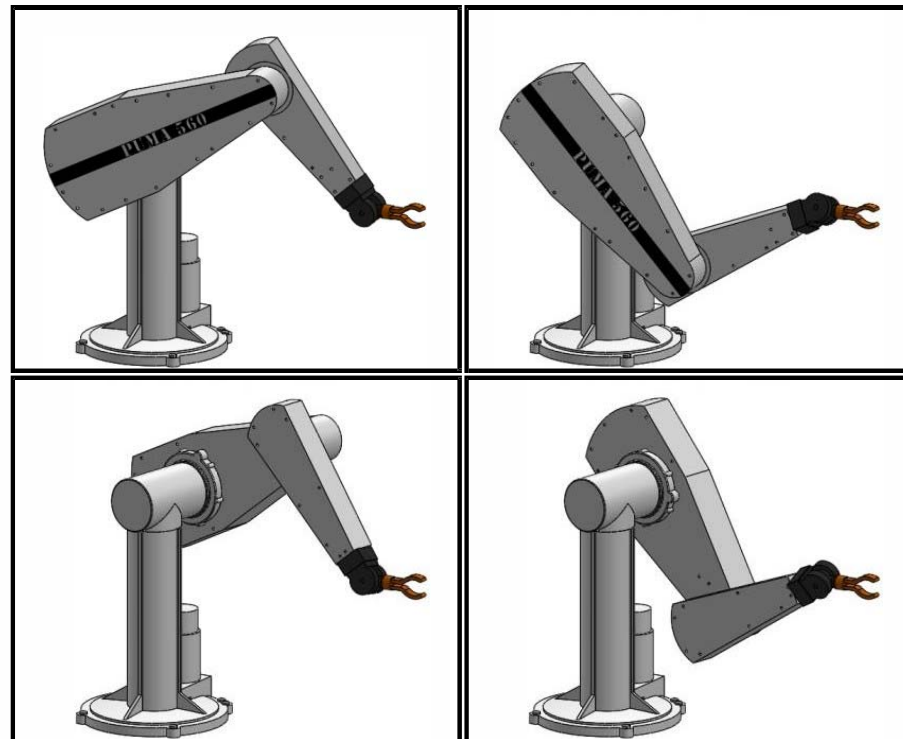
- ◆ 每一个姿态中，具有2组手腕转动姿态

$$\theta'_4 = \theta_4 + 180^\circ$$

$$\theta'_5 = -\theta_5$$

$$\theta'_6 = \theta_6 + 180^\circ$$

- ◆ 若手臂本身有几何限制，并非每一种解都可以运作



多重解 -3

□ 若具有多重解，解的选择方式

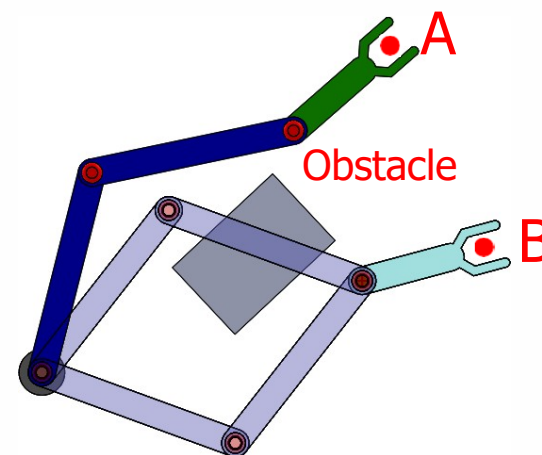
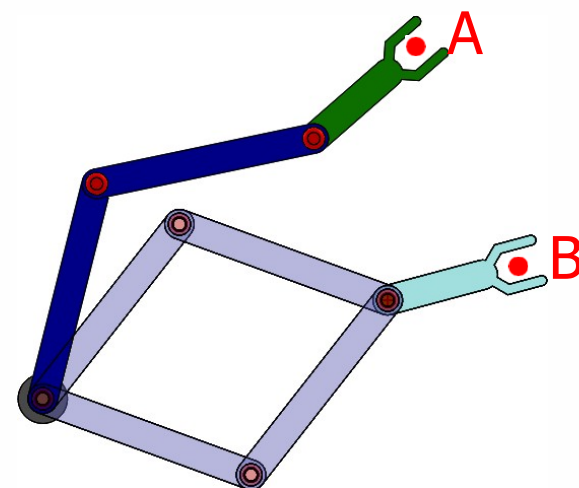
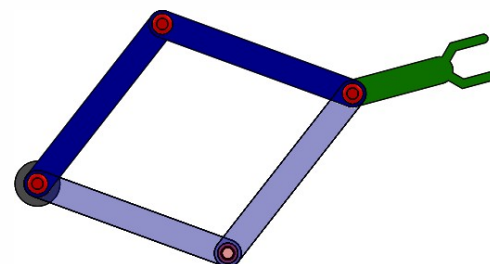
◆ 离目前状态最近的解

○ 最快

○ 最省能

○ ...

◆ 避开障碍物



求解方法

- 解析法 Closed-form solutions
 - ◆ 用 代数algebraic 或 几何geometric 方法
- 数值法 Numerical solutions
- 目前大多机械手臂设计成具有解析解
 - ◆ Pieper's solution: 相邻三轴相交一点

Example: A RRR Manipulator -1

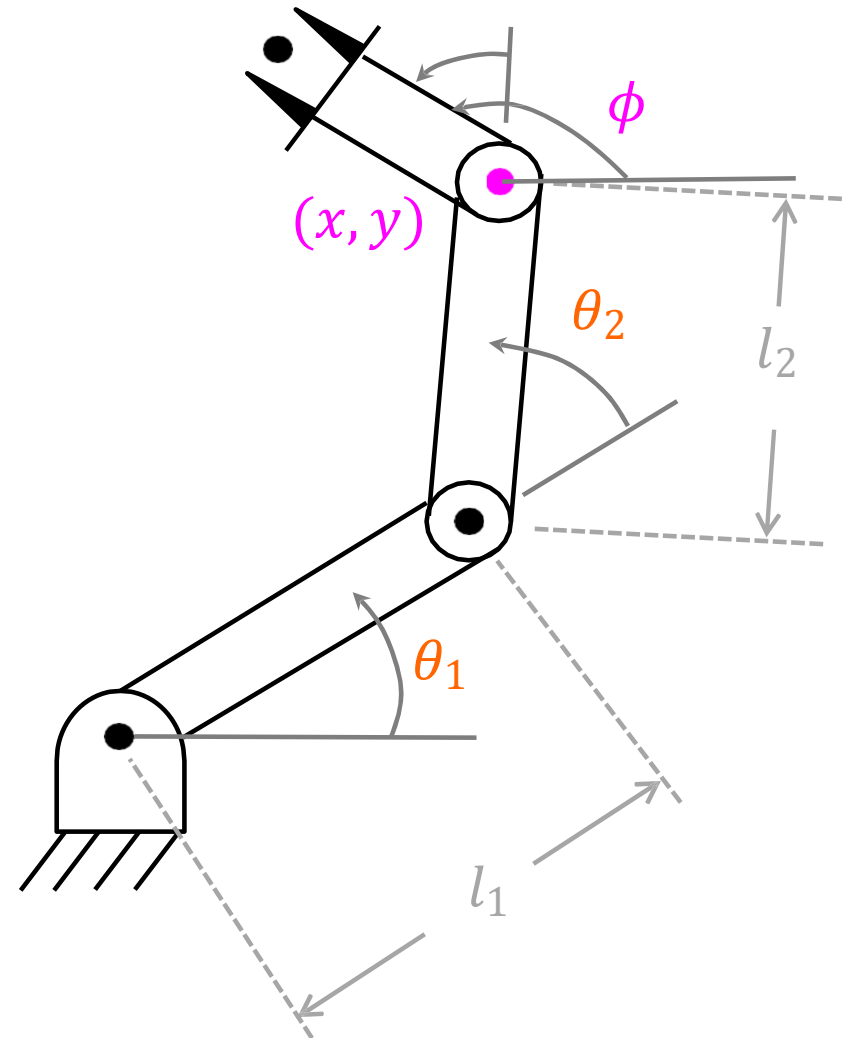
□ 给定: (x, y, ϕ) , 那么 $(\theta_1, \theta_2, \theta_3) = ?$

◆ 正运动学

$${}^0_3T = \begin{bmatrix} c_{123} & -s_{123} & 0.0 & l_1 c_1 + l_2 c_{12} \\ s_{123} & c_{123} & 0.0 & l_1 s_1 + l_2 s_{12} \\ 0.0 & 0.0 & 1.0 & 0.0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

◆ 目标点

$${}^0_3T = \begin{bmatrix} c_\phi & -s_\phi & 0.0 & x \\ s_\phi & c_\phi & 0.0 & y \\ 0.0 & 0.0 & 1.0 & 0.0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



Example: A RRR Manipulator -2

- 几何法：将空间几何切割成平面几何

$$x^2 + y^2 = l_1^2 + l_2^2 - 2l_1l_2\cos(180^\circ - \theta_2)$$

$$c_2 = \frac{x^2 + y^2 - l_1^2 - l_2^2}{2l_1l_2}$$

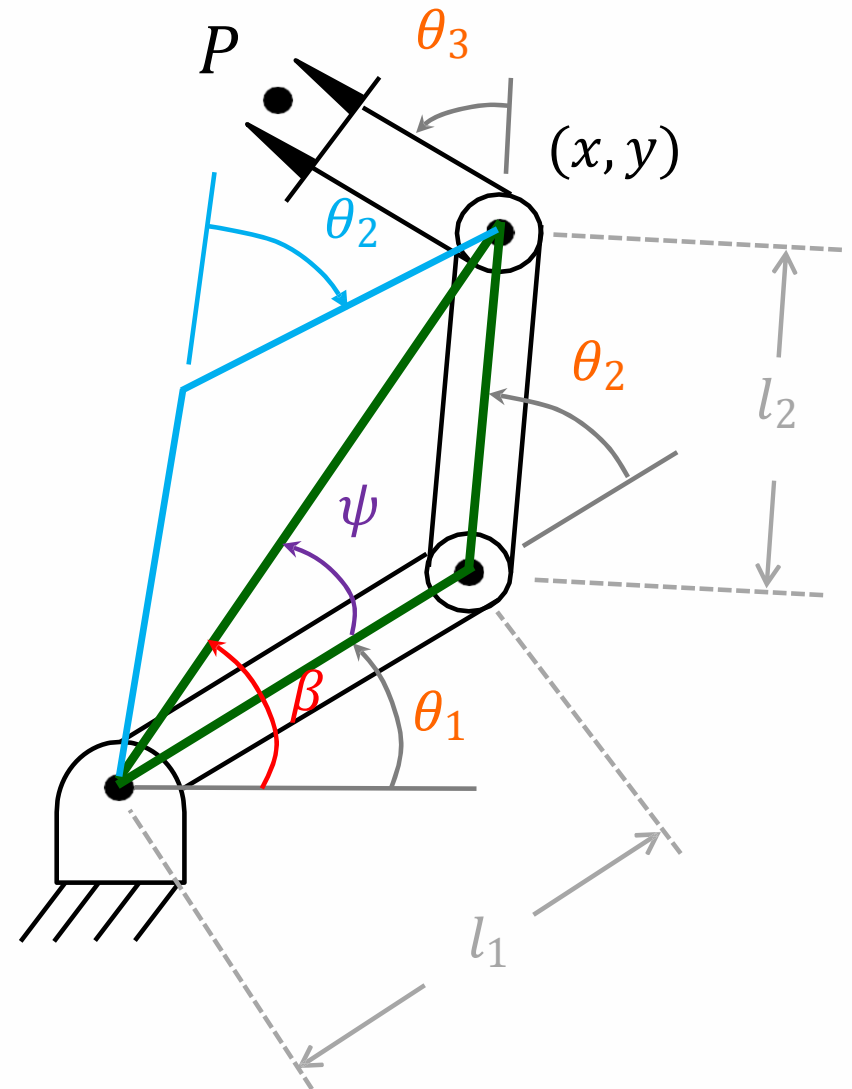
余弦定理

$$\cos\psi = \frac{l_2^2 - (x^2 + y^2) - l_1^2}{-2l_1\sqrt{x^2 + y^2}}$$

三角形内角 $0^\circ < \psi < 180^\circ$

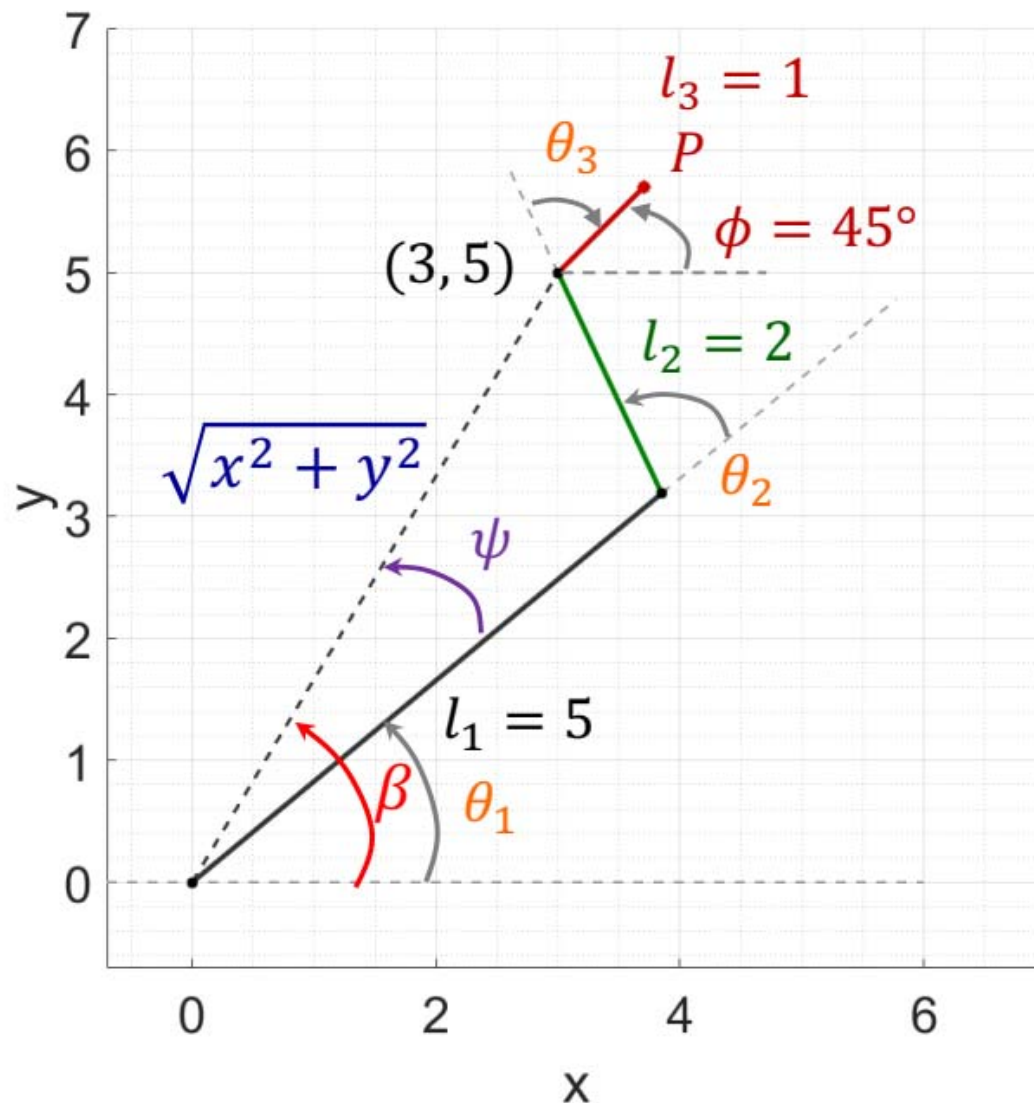
$$\theta_1 = \begin{cases} \text{atan2}(y, x) + \psi & \theta_2 < 0^\circ \\ \text{atan2}(y, x) - \psi & \theta_2 > 0^\circ \end{cases}$$

$$\theta_3 = \phi - \theta_1 - \theta_2$$



Example: A RRR Manipulator -3

□ Ex: 量化计算



$$c_2 = \frac{x^2 + y^2 - l_1^2 - l_2^2}{2l_1l_2}$$

$$\theta_2 = 75.5^\circ$$

$$\cos\psi = \frac{l_2^2 - (x^2 + y^2) - l_1^2}{-2l_1\sqrt{x^2 + y^2}}$$

$$\psi = 19.4^\circ$$

$$\theta_1 = \text{atan2}(y, x) - \psi$$

$$\theta_1 = 39.6^\circ$$

$$\theta_3 = \phi - \theta_1 - \theta_2$$

$$\theta_3 = -70.2^\circ$$

Example: A RRR Manipulator -5

□ 代数解

◆ 建立方程式

$$c_\phi = c_{123}$$

$$s_\phi = s_{123}$$

$$x = l_1 c_1 + l_2 c_{12}$$

$$y = l_1 s_1 + l_2 s_{12}$$

$$\begin{aligned} {}^0_3T &= \begin{bmatrix} c_{123} & -s_{123} & 0.0 & l_1 c_1 + l_2 c_{12} \\ s_{123} & c_{123} & 0.0 & l_1 s_1 + l_2 s_{12} \\ 0.0 & 0.0 & 1.0 & 0.0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} c_\phi & -s_\phi & 0.0 & x \\ s_\phi & c_\phi & 0.0 & y \\ 0.0 & 0.0 & 1.0 & 0.0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \end{aligned}$$

◆ 解 θ_2

$$x^2 + y^2 = l_1^2 + l_2^2 + 2l_1l_2c_2$$

$$c_2 = \frac{x^2 + y^2 - l_1^2 - l_2^2}{2l_1l_2}$$

> 1 or < -1: 太远以至于机械臂够不到

-1 ≤ ≤ 1: “两种解”

$$\theta_2 = \cos^{-1}(c_2)$$



Example: A RRR Manipulator

-6

- ◆ 将求得的 θ_2 带入方程式

$$x = l_1 c_1 + l_2 c_{12} = (l_1 + l_2 c_2) c_1 + (-l_2 s_2) s_1 \triangleq k_1 c_1 - k_2 s_1$$

$$y = l_1 s_1 + l_2 s_{12} = (l_1 + l_2 c_2) s_1 + (l_2 s_2) c_1 \triangleq k_1 s_1 + k_2 c_1$$

- ◆ 变数变换

定义

$$r = +\sqrt{k_1^2 + k_2^2}$$

且有

$$k_1 = r \cos \gamma$$

$$\gamma = \text{Atan2}(k_2, k_1)$$

$$k_2 = r \sin \gamma$$

得到

$$\frac{x}{r} = \cos \gamma \cos \theta_1 - \sin \gamma \sin \theta_1 = \cos(\gamma + \theta_1)$$

$$\frac{y}{r} = \cos \gamma \sin \theta_1 + \sin \gamma \cos \theta_1 = \sin(\gamma + \theta_1)$$

Example: A RRR Manipulator -7

◆ 解 θ_1

$$\gamma + \theta_1 = \text{Atan2}\left(\frac{y}{r}, \frac{x}{r}\right) = \text{Atan2}(y, x)$$

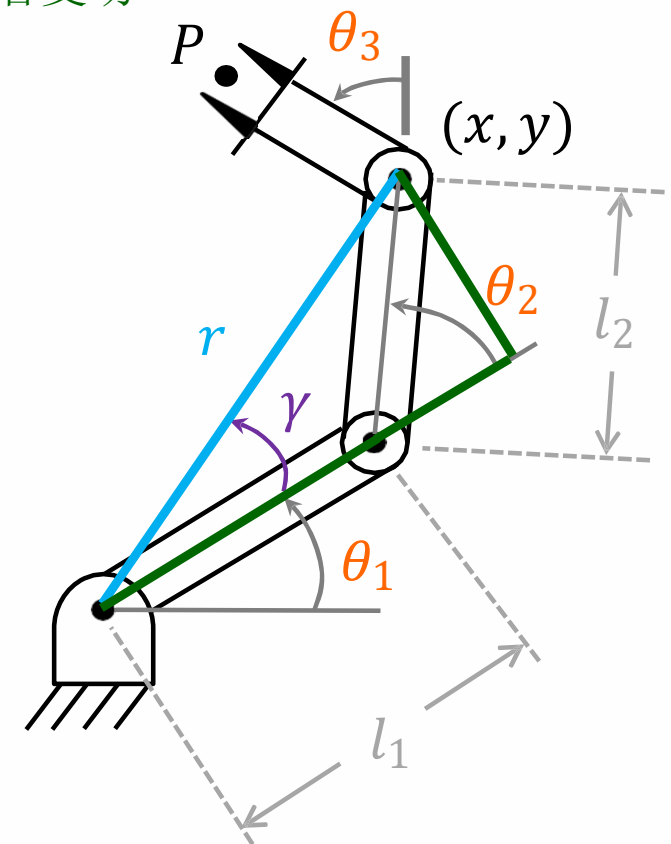
➡ $\theta_1 = \text{Atan2}(y, x) - \text{Atan2}(k_2, k_1)$

当 θ_2 选不同解, c_2 和 s_2 变动, k_1 和 k_2 变动, θ_1 也跟者变动

◆ 解 θ_3

$$\theta_1 + \theta_2 + \theta_3 = \text{Atan2}(s_\phi, c_\phi) = \phi$$

➡ $\theta_3 = \phi - \theta_1 - \theta_2$



三角函数方程式求解

□ Ex: 如何求得 $a\cos\theta + b\sin\theta = c$ 的 θ ?

◆ 方法: 变换到多项式 (4阶以下有解析解)

$$\tan\left(\frac{\theta}{2}\right) = u, \quad \cos\theta = \frac{1-u^2}{1+u^2}, \quad \sin\theta = \frac{2u}{1+u^2}$$

◆ 步骤:

$$a\cos\theta + b\sin\theta = c$$

$$a \frac{1-u^2}{1+u^2} + b \frac{2u}{1+u^2} = c$$

$$(a+c)u^2 - 2bu + (c-a) = 0$$

$$u = \frac{b \pm \sqrt{b^2 + a^2 - c^2}}{a+c}$$

a, b, c大小有限制, 不一定有解

$$\theta = 2 \tan^{-1}\left(\frac{b \pm \sqrt{b^2 + a^2 - c^2}}{a+c}\right) \quad a+c \neq 0$$

$$\theta = 180^\circ \quad a+c = 0$$

Pieper's Solution -1

□ 若6-DOF manipulator具有三个连续的轴交 在同一点，
则手臂有解析解

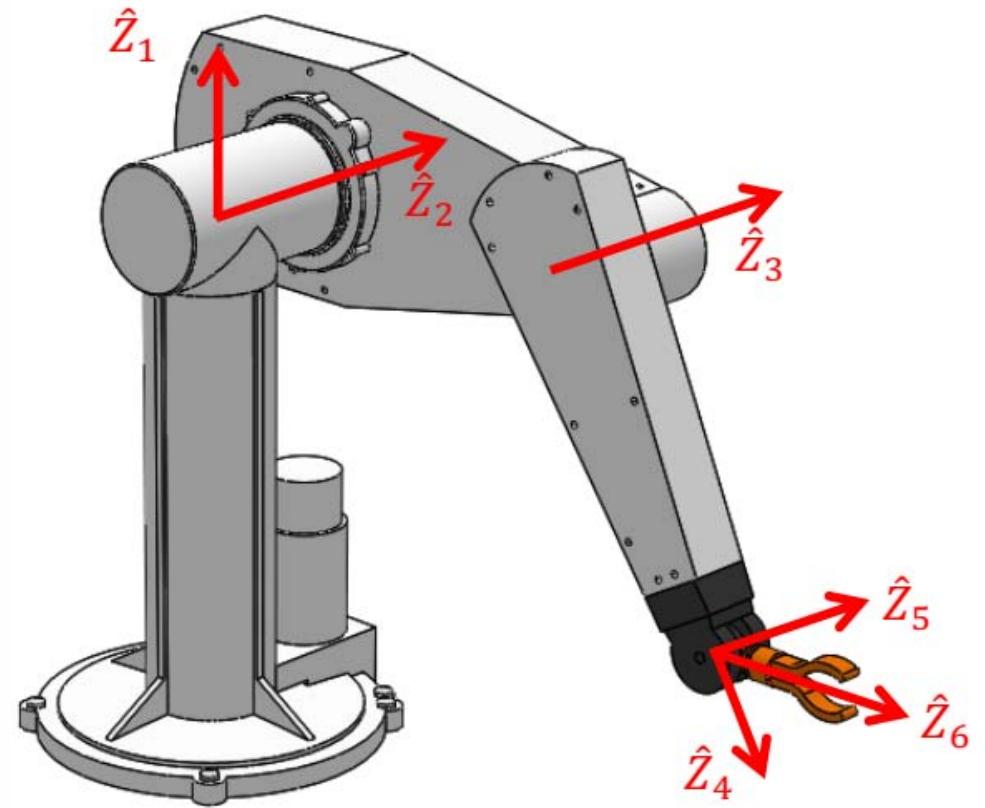
□ 一般，会把后三轴如此设计

- ◆ 前三轴：产生移动
- ◆ 后三轴：产生转动

□ Ex: A RRRRRR manipulator

- ◆ 因后三轴交一点

$${}^0P_{6\text{ ORG}} = {}^0P_{4\text{ ORG}}$$



Pieper's Solution -2

□ Positioning structure

◆ 法则：让 $\theta_1, \theta_2, \theta_3$ 层层分离

$$\begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = {}^0P_{4ORG} = {}^0T_1 {}^1T_2 {}^2T_3 {}^3P_{4ORG}$$

Note: ${}^{i-1}T_i = \begin{bmatrix} c\theta_i & -s\theta_i & 0 & a_{i-1} \\ s\theta_i c\alpha_{i-1} & c\theta_i c\alpha_{i-1} & -s\alpha_{i-1} & -s\alpha_{i-1}d_i \\ s\theta_i s\alpha_{i-1} & c\theta_i s\alpha_{i-1} & c\alpha_{i-1} & c\alpha_{i-1}d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$

$$= {}^0T_1 {}^1T_2 {}^2T_3 \begin{bmatrix} a_3 \\ -d_4 s\alpha_3 \\ d_4 c\alpha_3 \\ 1 \end{bmatrix} = {}^0T_1 {}^1T_2 \begin{bmatrix} f_1(\theta_3) \\ f_2(\theta_3) \\ f_3(\theta_3) \\ 1 \end{bmatrix}$$

4th column of 3T_4

$\cos \theta_i = c\theta_i = c_i$
 $\sin \theta_i = s\theta_i = s_i$

so

$$\begin{bmatrix} f_1(\theta_3) \\ f_2(\theta_3) \\ f_3(\theta_3) \\ 1 \end{bmatrix} = {}^2T_3 \begin{bmatrix} a_3 \\ -d_4 s\alpha_3 \\ d_4 c\alpha_3 \\ 1 \end{bmatrix}$$

让 $\theta_1, \theta_2, \theta_3$ 层层分离, f 为 θ_3 函数

$$f_1(\theta_3) = a_3 c_3 + d_4 s\alpha_3 s_3 + a_2$$

$$f_2(\theta_3) = a_3 c\alpha_2 s_3 - d_4 s\alpha_3 c\alpha_2 c_3 - d_4 s\alpha_2 c\alpha_3 - d_3 s\alpha_2$$

$$f_3(\theta_3) = a_3 s\alpha_2 s_3 - d_4 s\alpha_3 s\alpha_2 c_3 + d_4 c\alpha_2 c\alpha_3 + d_3 c\alpha_2$$

Pieper's Solution -3

◆ 下一步

$${}^0P_{4ORG} = \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = {}^0_1T {}^1_2T \begin{bmatrix} f_1(\theta_3) \\ f_2(\theta_3) \\ f_3(\theta_3) \\ 1 \end{bmatrix} = {}^0_1T \begin{bmatrix} g_1(\theta_2, \theta_3) \\ g_2(\theta_2, \theta_3) \\ g_3(\theta_2, \theta_3) \\ 1 \end{bmatrix} = \begin{bmatrix} c_1g_1 - s_1g_2 \\ s_1g_1 + c_1g_2 \\ g_3 \\ 1 \end{bmatrix}$$

让 $\theta_1, \theta_2, \theta_3$ 层层分离, g 为 θ_2, θ_3 函数

$$g_1(\theta_2, \theta_3) = c_2f_1 - s_2f_2 + a_1$$

$$g_2(\theta_2, \theta_3) = s_2c\alpha_1f_1 + c_2c\alpha_1f_2 - s\alpha_1f_3 - d_2s\alpha_1$$

$$g_3(\theta_2, \theta_3) = s_2s\alpha_1f_1 + c_2s\alpha_1f_2 + c\alpha_1f_3 + d_2c\alpha_1$$

$$\begin{aligned} r &= x^2 + y^2 + z^2 = g_1^2 + g_2^2 + g_3^2 && r \text{ 仅为 } \theta_2, \theta_3 \text{ 函数} \\ &= f_1^2 + f_2^2 + f_3^2 + a_1^2 + d_2^2 + 2d_2f_3 + 2a_1(c_2f_1 - s_2f_2) \\ &= (k_1c_2 + k_2s_2)2a_1 + k_3 \end{aligned}$$

$$k_1(\theta_3) = f_1$$

$$k_2(\theta_3) = -f_2$$

$$k_3(\theta_3) = f_1^2 + f_2^2 + f_3^2 + a_1^2 + d_2^2 + 2d_2f_3$$

Pieper's Solution -4

◆ 此外

$$z = g_3 = (k_1 s_2 - k_2 c_2) s \alpha_1 + k_4$$

z 仅为 θ_2, θ_3 函数

$$k_1(\theta_3) = f_1$$

$$k_2(\theta_3) = -f_2$$

$$k_4(\theta_3) = f_3 c \alpha_1 + d_2 c \alpha_1$$

◆ 整合 r 和 z 一起考虑

$$\begin{cases} r = (k_1 c_2 + k_2 s_2) 2a_1 + k_3 \\ z = (k_1 s_2 - k_2 c_2) s \alpha_1 + k_4 \end{cases}$$

◦ If $a_1 = 0$, $r = k_3(\theta_3) = f_1^2 + f_2^2 + f_3^2 + a_1^2 + d_2^2 + 2d_2 f_3$

◦ If $s \alpha_1 = 0$, $z = k_4(\theta_3) = f_3 c \alpha_1 + d_2 c \alpha_1$

◦ Else

$$\frac{(r - k_3)^2}{4a_1^2} + \frac{(z - k_4)^2}{s^2 \alpha_1} = k_1^2 + k_2^2$$



Solve θ_3 of all three cases by using " $u = \tan\left(\frac{\theta_3}{2}\right)$ "

Pieper's Solution -5

□ 最后

Using $r = (k_1 c_2 + k_2 s_2)2a_1 + k_3$ to solve θ_2

Using $x = c_1 g_1(\theta_2, \theta_3) - s_1 g_2(\theta_2, \theta_3)$ to solve θ_1

□ 姿态

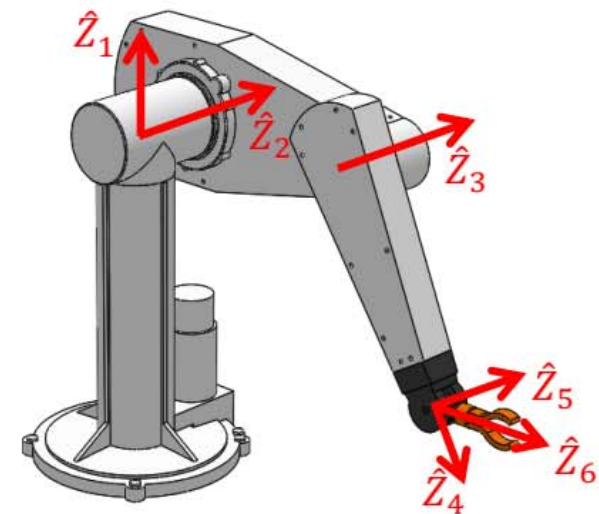
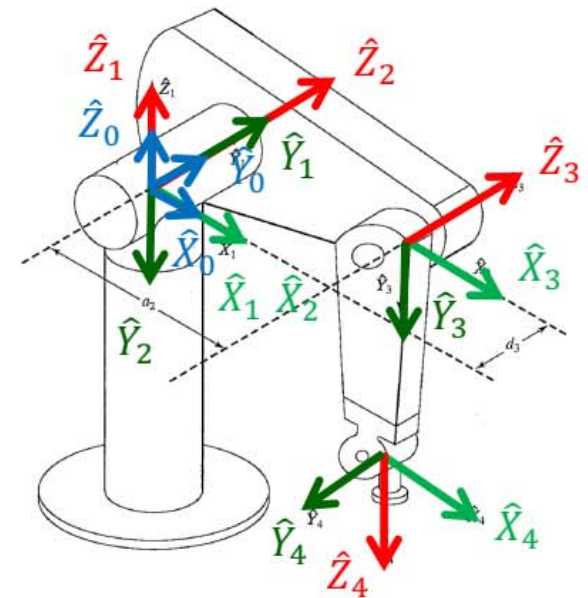
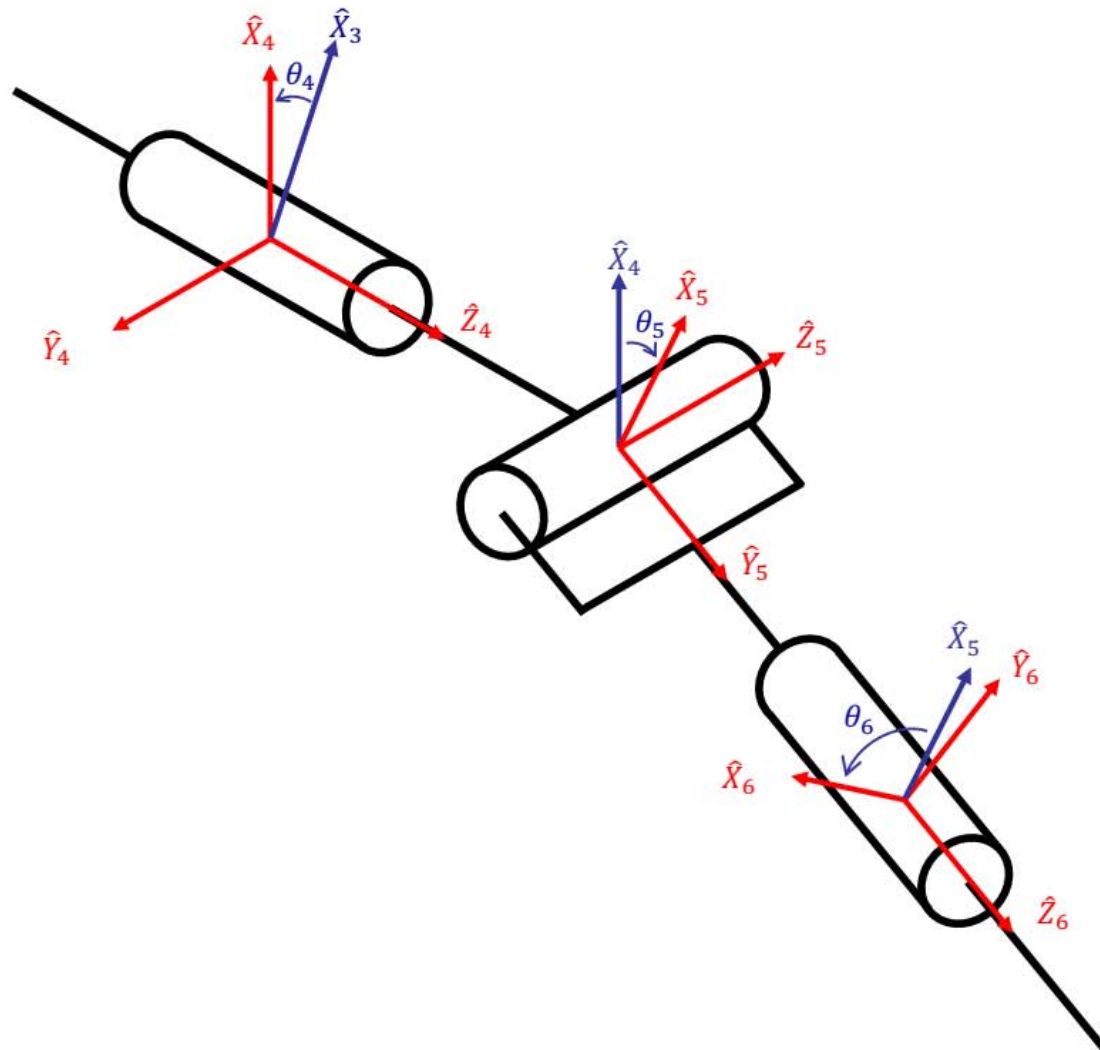
◆ $\theta_1, \theta_2, \theta_3$ 已知

$${}^3_6R = {}^0_3R^{-1} {}^10_6R$$

◆ 以 Z-Y-Z Euler angle 求解 $\theta_4, \theta_5, \theta_6$

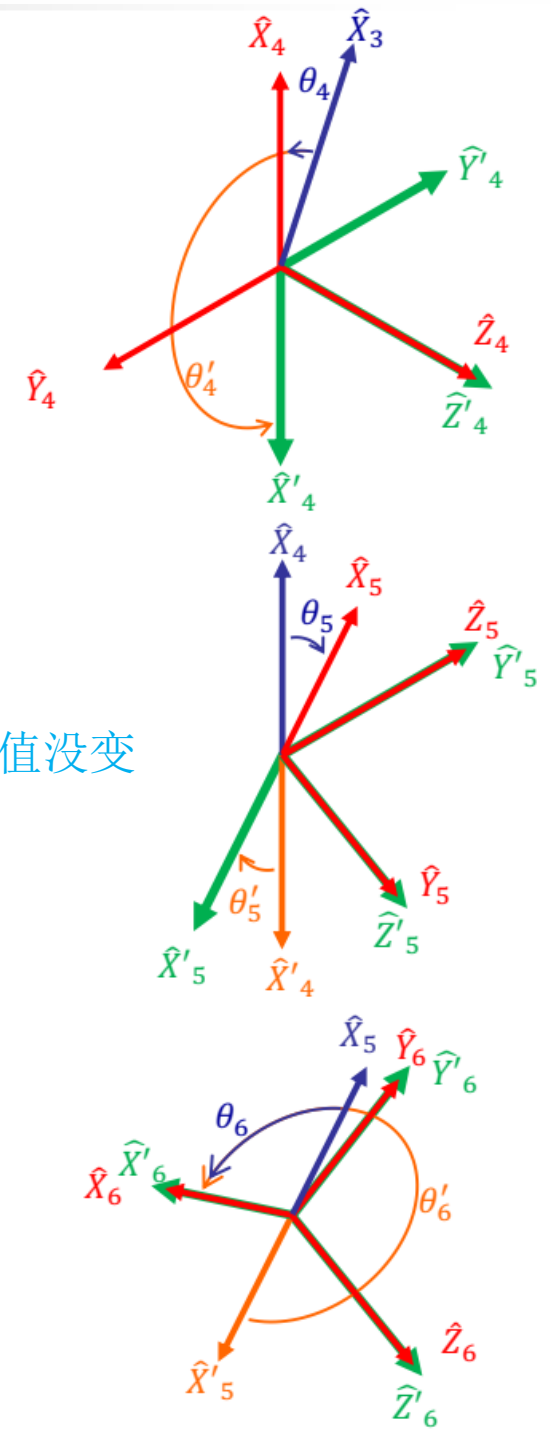
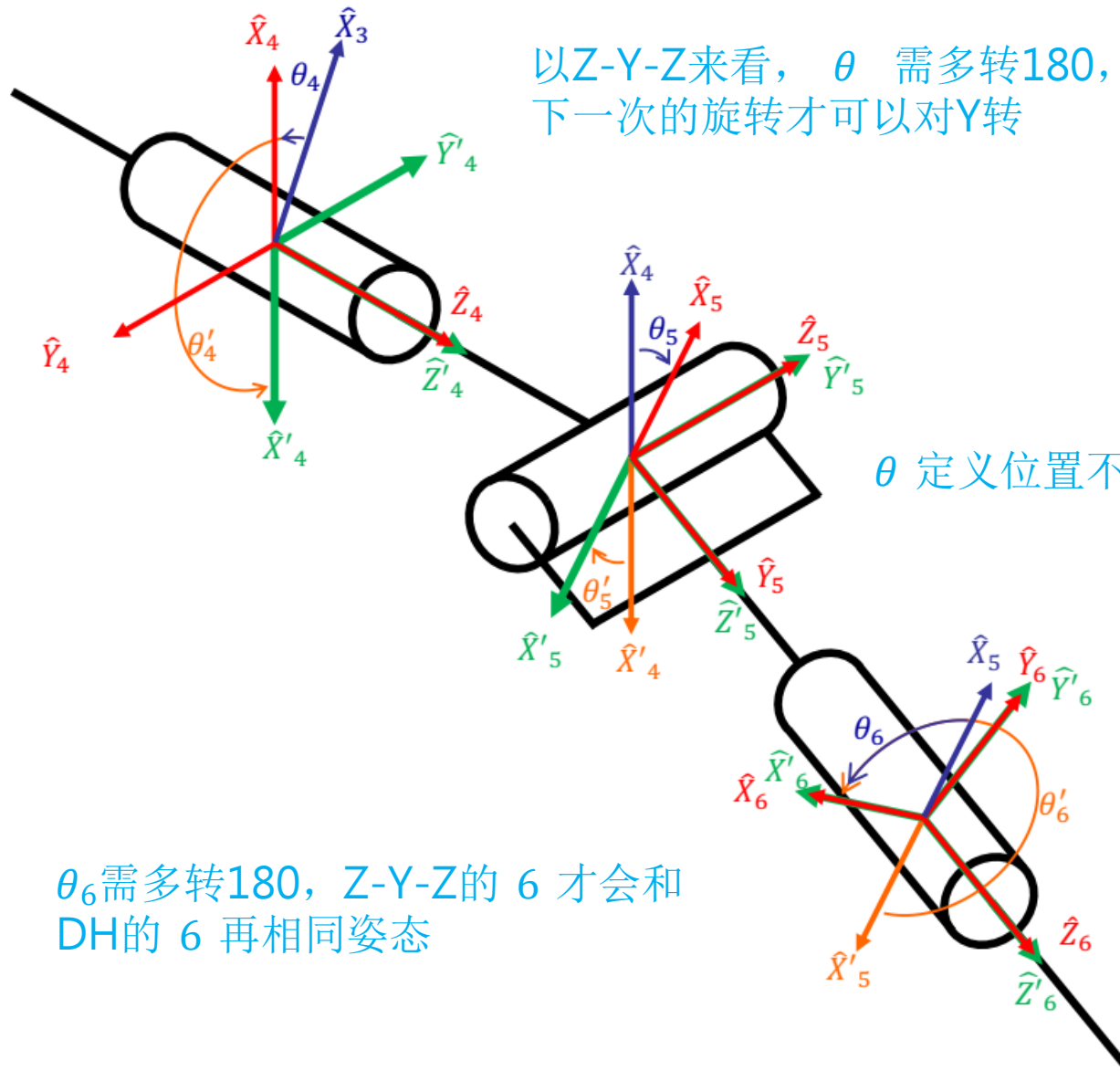
Pieper's Solution -6

□ Joints 4-6, DH definition



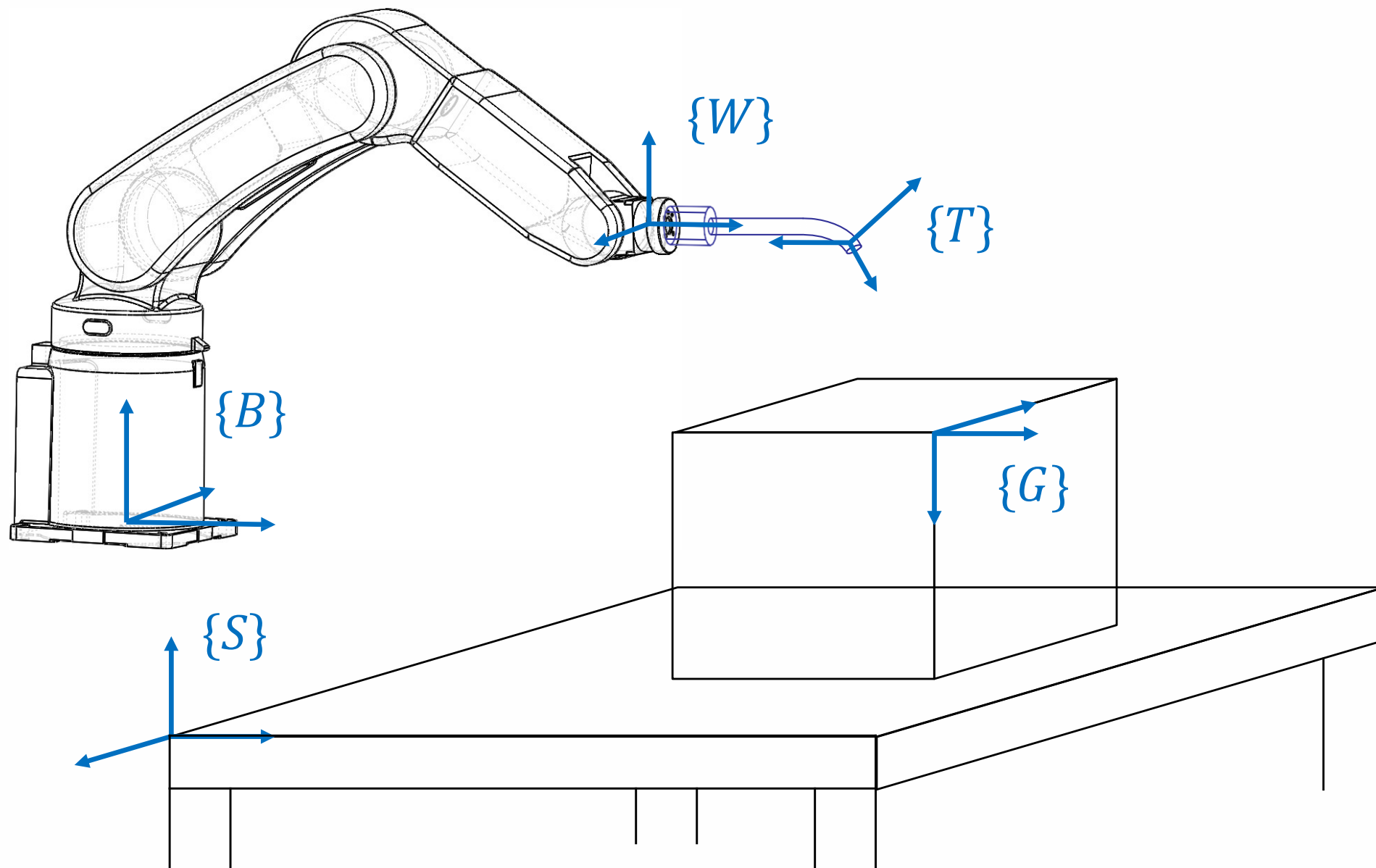
Pieper's Solution -7

□ DH definition vs. Z-Y-Z Euler Angles



坐标系

- Base, wrist, tool, station, and goal frames

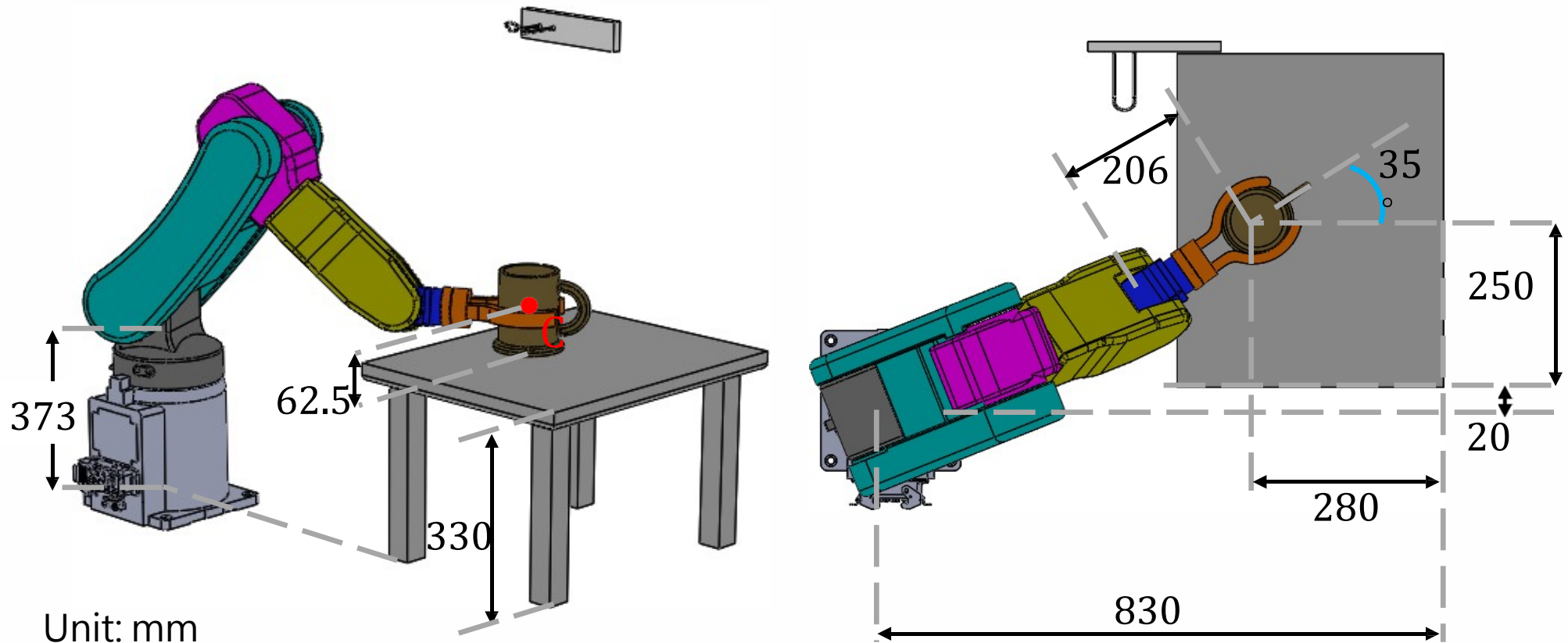


Example: 物件取放任务 -1

- 情境：机械手臂夹住放在桌上的杯子，移动手臂将杯子挂到墙上的杯架

Example: 物件取放任务 -2

- 现阶段任务：为使RRRRRRR手臂能以下图姿态夹住杯子（任务的起始点C），手臂的6个joint angles需为何？

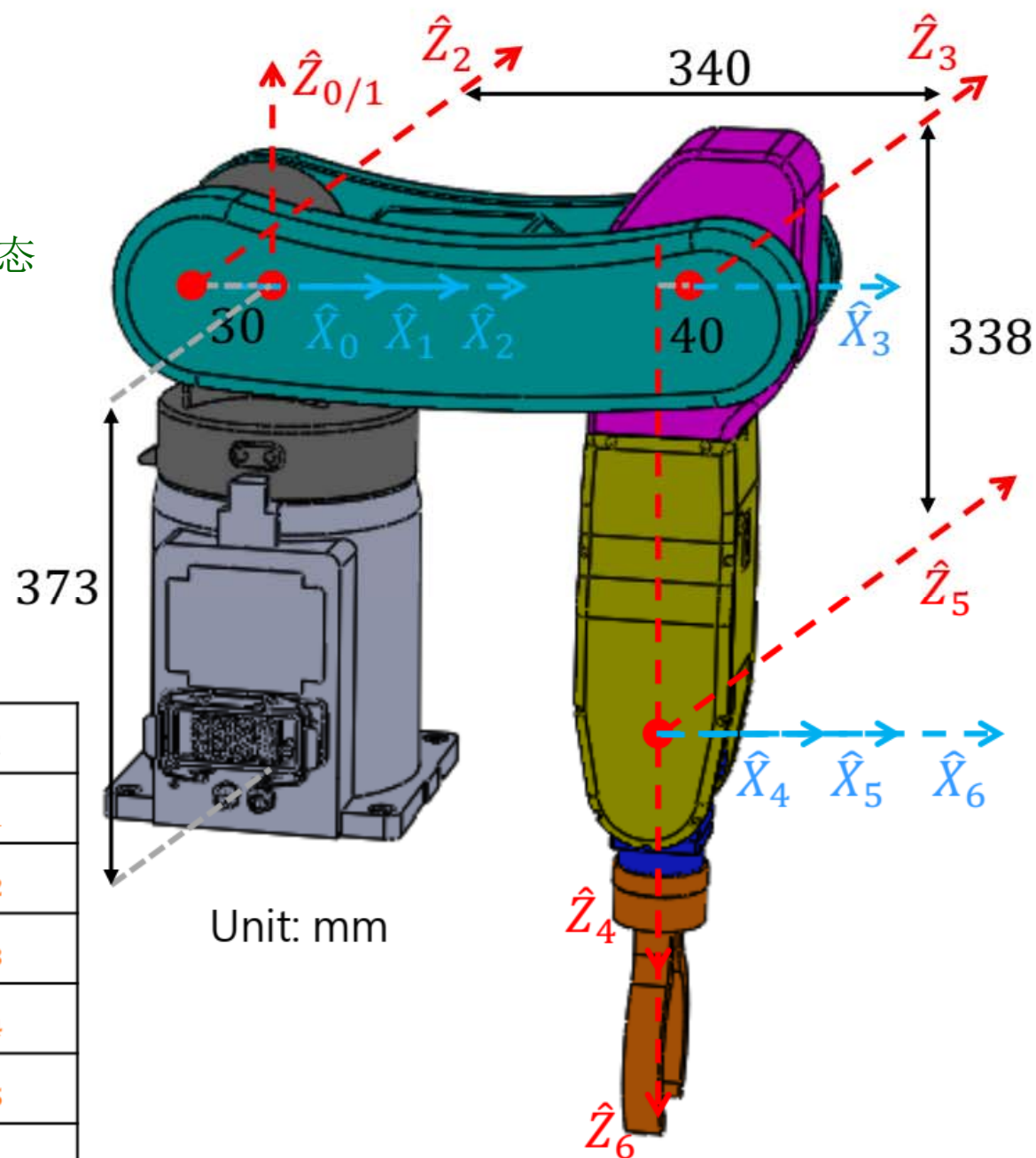


Example: 物件取放任务 -3

□ Step 1: 定义DH Table

图中显示各轴为 0° 的状态

i	α_{i-1}	a_{i-1}	d_i	θ_i
1	0°	0	0	θ_1
2	-90°	$a_1 = -30$	0	θ_2
3	0°	$a_2 = 340$	0	θ_3
4	-90°	$a_3 = -40$	$d_4 = 338$	θ_4
5	90°	0	0	θ_5
6	-90°	0	0	θ_6

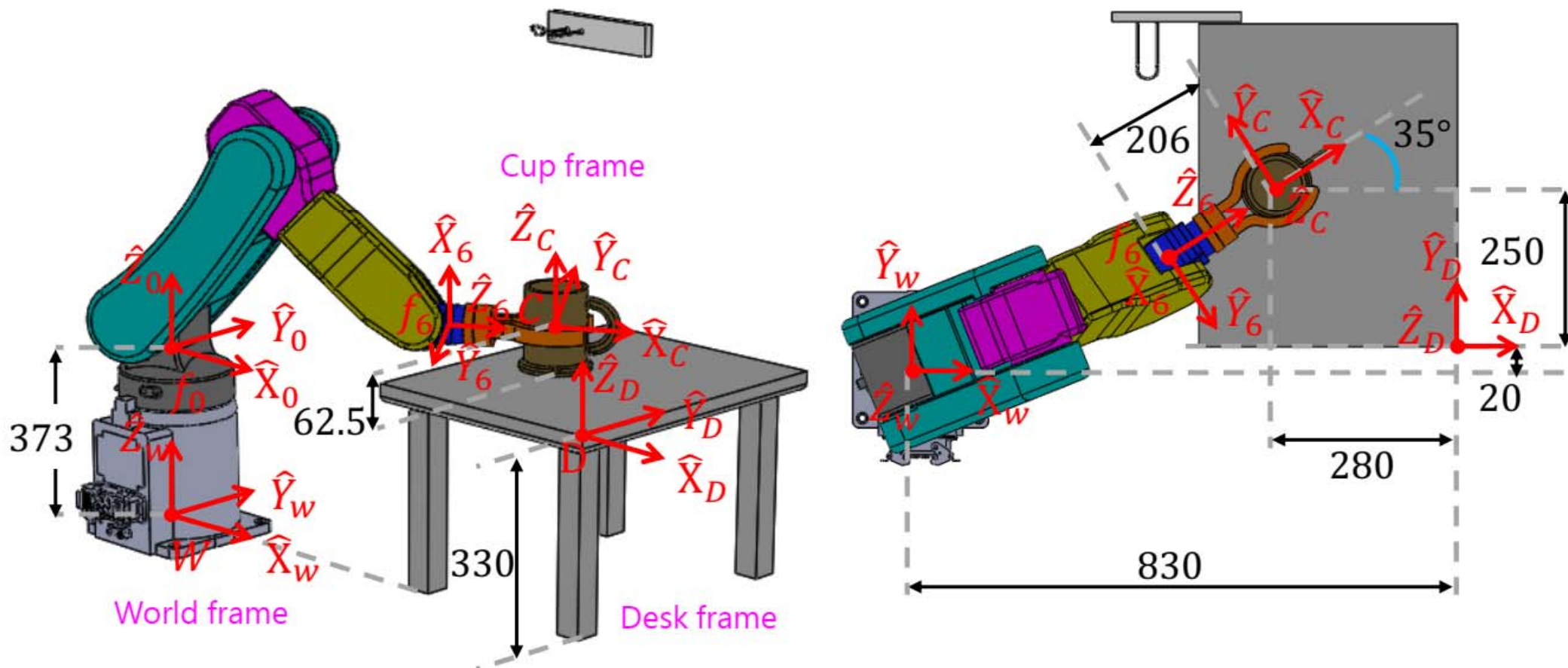


Example: 物件取放任务 -4

- Step 2: 找出 ${}^W T_C$, 再进一步找出 ${}^0 T_6$

$${}^W T_C \equiv {}^W T_D {}^D T_C \equiv \begin{bmatrix} 1 & 0 & 0 & 830 \\ 0 & 1 & 0 & 20 \\ 0 & 0 & 1 & 330 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos 35^\circ & -\sin 35^\circ & 0 & -280 \\ \sin 35^\circ & \cos 35^\circ & 0 & 250 \\ 0 & 0 & 1 & 62.5 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

由「桌子相对于手臂」和「杯子相对于桌子」的相对关系推得



Example: 物件取放任务 -5

$${}^W_c T = {}^W_0 T {}^0_6 T {}^6_c T$$

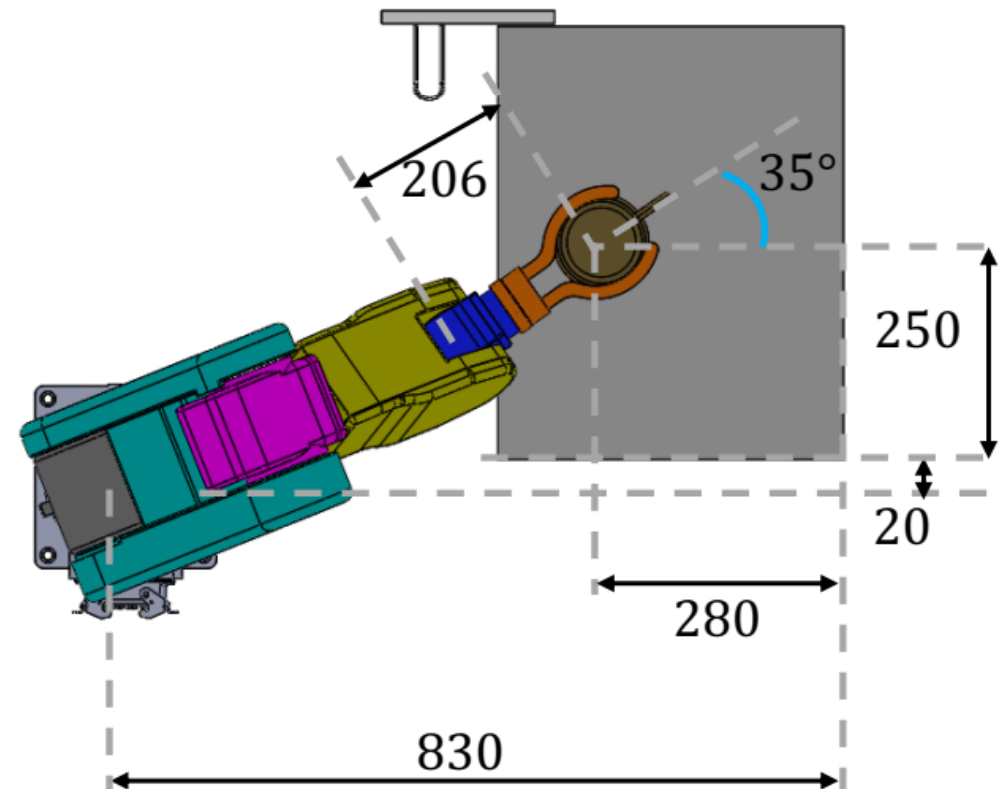
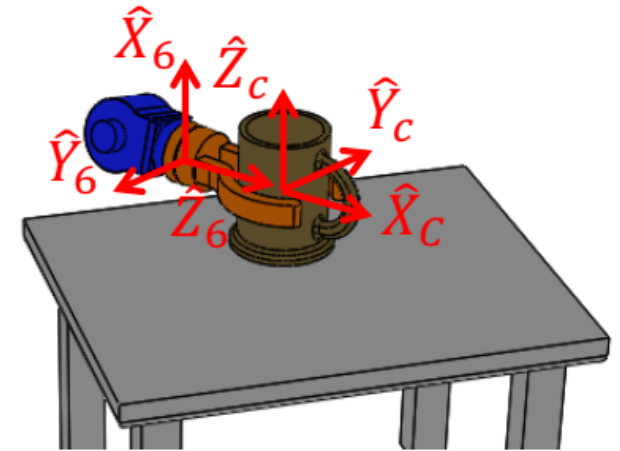
$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 373 \\ 0 & 0 & 0 & 1 \end{bmatrix} {}^0_6 T \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 206 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^0_6 T = {}^W_0 T^{-1} {}^W_c T {}^6_c T^{-1}$$

$$= \begin{bmatrix} 0 & 0.5736 & 0.8192 & 381.3 \\ 0 & -0.8192 & 0.5736 & 151.8 \\ 1 & 0 & 0 & 19.5 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^0_6 R = \begin{bmatrix} 0 & 0.5736 & 0.8192 \\ 0 & -0.8192 & 0.5736 \\ 1 & 0 & 0 \end{bmatrix}$$

$${}^0 P_{6 ORG} = \begin{bmatrix} 381.3 \\ 151.8 \\ 19.5 \end{bmatrix}$$



Example: 物件取放任务 -6

□ Step 3: 找出 $\theta_1 - \theta_6$

◆ $\theta_1 \theta_2 \theta_3$ 角度求解

$$\begin{bmatrix} f_1(\theta_3) \\ f_2(\theta_3) \\ f_3(\theta_3) \\ 1 \end{bmatrix} = {}^2_3T {}^3P_{4ORG}$$
$$= \begin{bmatrix} c_3 & -s_3 & 0 & 340 \\ s_3 & c_3 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -40 \\ 338 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 340 - 338s_3 - 40c_3 \\ 338c_3 - 40s_3 \\ 0 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} g_1(\theta_2, \theta_3) \\ g_2(\theta_2, \theta_3) \\ g_3(\theta_2, \theta_3) \\ 1 \end{bmatrix} = {}^1_2T \begin{bmatrix} f_1(\theta_3) \\ f_2(\theta_3) \\ f_3(\theta_3) \\ 1 \end{bmatrix}$$
$$= \begin{bmatrix} c_2 & -s_2 & 0 & -30 \\ 0 & 0 & 1 & 0 \\ -s_2 & -c_2 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} f_1(\theta_3) \\ f_2(\theta_3) \\ f_3(\theta_3) \\ 1 \end{bmatrix} = \begin{bmatrix} 340c_2 - 40c_{23} - 338s_{23} - 30 \\ 0 \\ 40s_{23} - 338c_{23} - 340s_2 \\ 1 \end{bmatrix}$$

Example: 物件取放任务 -7

$$r = (k_1 c_2 + k_2 s_2) 2a_1 + k_3 = ||P||^2 = 168813.18$$

$$z = (k_1 s_2 - k_2 c_2) s \alpha_1 + k_4 = 19.5$$

计算 θ_1 θ_2 θ_3 角度

$$\frac{(r-k_3)^2}{4a_1^2} + \frac{(z-k_4)^2}{s^2 \alpha_1^2} = k_1^2 + k_2^2$$

$$\Rightarrow \text{solve } \theta_3 = 2.5^\circ$$

$$r = (k_1 c_2 + k_2 s_2) 2a_1 + k_3$$

$$\Rightarrow \text{solve } \theta_2 = -52.2^\circ$$

$$x = c_1 g_1(\theta_2, \theta_3) - s_1 g_2(\theta_2, \theta_3)$$

$$\Rightarrow \text{solve } \theta_1 = 21.8^\circ$$

Example: 物件取放任务 -8

◆ $\theta_4 \theta_5 \theta_6$ 角度求解

$${}^0_3R = \begin{bmatrix} 0.6006 & 0.7082 & -0.3710 \\ 0.24 & 0.2830 & 0.9286 \\ 0.7627 & -0.6468 & 0 \end{bmatrix}$$

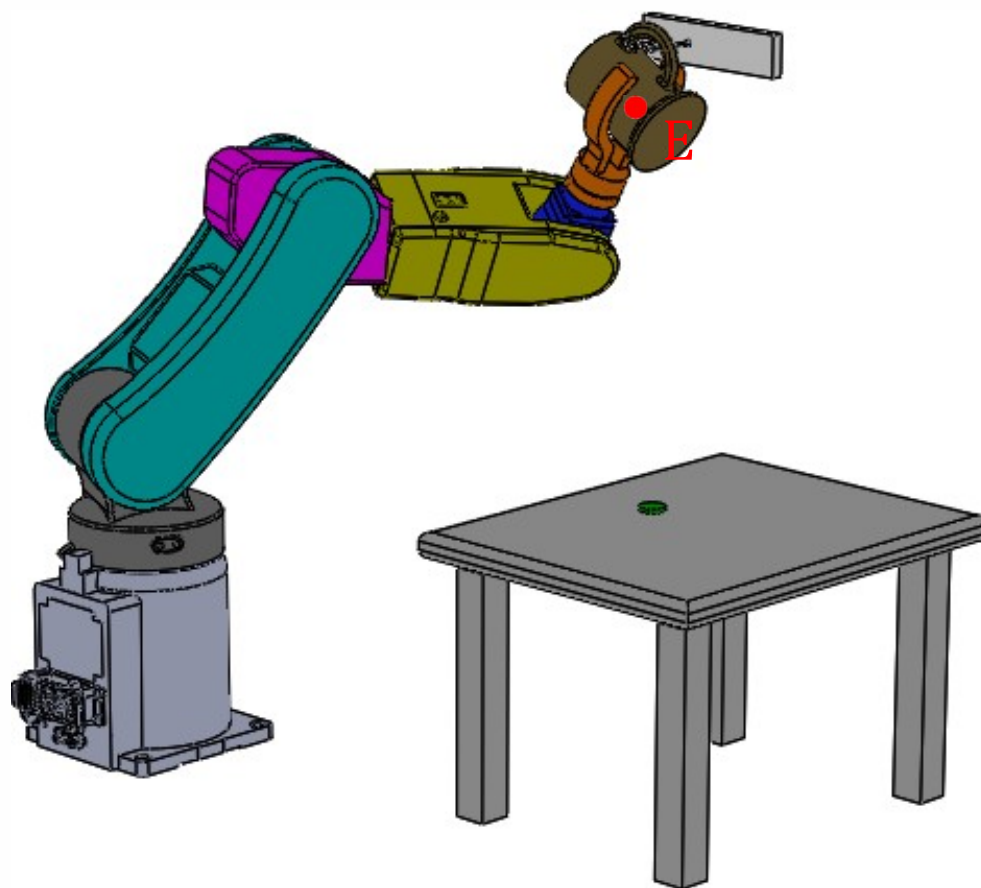
$${}^3_6R = {}^0_3R^{-1} {}^0_6R = \begin{bmatrix} 0.7627 & 0.1477 & 0.6297 \\ -0.6468 & 0.1744 & 0.7424 \\ 0 & -0.9735 & 0.2286 \end{bmatrix}$$

使用Z-Y-Z Euler angle求得剩下的joint angles

$$\theta_4 = -20^\circ \quad \theta_5 = -42^\circ \quad \theta_6 = 15^\circ$$

Example: 物件取放任务 -8

- Class Exercise: 要让夹爪到杯子的末端点E，6个转轴的转角又分别是？



后续课程安排

- 下周半期考
 - 半开卷: 一张A4纸的资料, 正反面写都行
- 期末考试用作业来复习
- 判断自由度
- 建立坐标系
- 建立DH表
- 画图-计算

考核成绩

考核要求:		期末占比
期末考试;		
笔试: 理论为主, 计算题为辅		50 %
半期考: 第 8 周		10 %
课程论文: 小学期交		20 %
作业: 课程作业 (前半程 4 次)		15 %
出勤: 扫码签到		5 %

课程论文Essay

国内多自由度工业操作臂的研究现状

- 1: 减速机
- 2: 电机
- 3: 控制器
- 4: 位置反馈装置
- 5: 夹持器或整机
- 6: 并联操作臂

要求:

字数>1K, 有参考文献, 有对比, 有思考
有观点。